

6-Week Internship Plan — Web Intern (Angular + .NET)

Project

WorkSync — Workforce & Task Management System

Project Overview

The company currently manages:

- employee attendance,
- task assignments,
- productivity tracking,
- and reporting

manually using Excel sheets and WhatsApp communication.

The goal is to build a centralized Workforce Management System where:

- Employees can log in
- Employees can mark attendance
- Managers can assign tasks
- Employees can update task progress
- Admins can manage users/departments
- Productivity dashboards and reports can be viewed

Web Technology Stack

Frontend

- Angular
- Angular Material
- RxJS
- TypeScript

Backend

- ASP.NET Core Web API
- Entity Framework Core
- SQL Server

Tools

- Git + GitHub
 - Swagger
 - Postman
 - VS Code / Visual Studio
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INTERN EXPECTATION

The intern is expected to:

- Build Angular screens
 - Integrate APIs
 - Build backend APIs
 - Design DB tables
 - Handle validations
 - Debug frontend/backend issues
 - Follow Git workflow
 - Present weekly progress
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Learning Plan (Week 1)

Angular Fundamentals

- What is Angular?
- Angular Architecture
- Angular CLI
- Components & Modules
- Data Binding
- Directives
- Routing Basics
- Services
- Dependency Injection
- Reactive Forms
- Form Validation

- HTTP Client
- Angular Material Introduction

Shared .NET Learning Topics

- What is ASP.NET Core?
- Web API Architecture
- Controllers & Routing
- HTTP Methods
- CRUD APIs
- Entity Framework Core
- SQL Basics
- Database Relationships
- DTO Basics
- API Testing using Swagger/Postman

Git Learning

- Git Branching
- Commit Standards
- Pull Requests

Assignment 1

Problem Statement

Build an Employee Management Module to replace manual employee tracking.

Frontend Requirements

Employee Listing Page

- Employee table/grid
- Search functionality
- Edit/Delete actions
- Status indicators

Add/Edit Employee Form

- Required field validation
- Email validation

- Validation messages
- Submit/cancel actions

Routing

- Employee listing route
- Add employee route
- Edit employee route

UI Expectations

- Responsive layout
- Loading indicators
- Snackbar/toast messages

Backend Requirements

APIs

- Get employees
- Get employee by ID
- Create employee
- Update employee
- Delete employee

Database

- Employee table
- Department table
- Relationships

Validation

- Duplicate email validation
- Required field validation

Deliverables

- Working CRUD module
 - API integration completed
 - Database configured
 - APIs tested in Postman
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Learning Plan (Week 2)

Angular Topics

- Lifecycle Hooks
- Route Guards
- Interceptors
- Shared Modules
- RxJS Basics
- Observables
- State Handling Basics

Shared .NET Learning Topics

- JWT Authentication
- Middleware
- Authorization
- Password Hashing
- Async/Await
- LINQ Basics

Assignment 2

Problem Statement

Build Authentication & Authorization Module.

Frontend Requirements

Login Screen

- Email/password validation
- Error handling
- Loading states
- Session persistence

Layout Structure

- Sidebar
- Header

- Logout functionality
- Role-based menus

Route Protection

- Protect unauthorized access
- Handle token expiration

API Interceptor

- Attach JWT token
- Handle global API errors

Backend Requirements

APIs

- Login API
- User validation API

JWT

- Token generation
- Role claims
- Expiration handling

Database

- Users table
- Roles table
- UserRoles table

Deliverables

- Secure login flow
- Protected routes
- JWT-secured APIs

Learning Plan (Week 3)

Angular Topics

- Parent-Child Communication

- Reusable Components
- Pipes
- Reactive Forms Advanced
- Dynamic Rendering
- Shared Components

Shared .NET Learning Topics

- Service Layer
- Repository Pattern Basics
- DTO Mapping
- API Validation
- Business Logic Separation

Assignment 3

Problem Statement

Build Attendance & Task Management Workflow.

Attendance Module

Features

- Mark attendance
- Attendance history
- Status indicators

Task Module

Manager Features

- Create task
- Assign employee
- Set due date
- Set priority

Employee Features

- View assigned tasks
- Update task status

- Add comments

Frontend Expectations

Task Listing

- Task cards/table
- Filters
- Status badges
- Priority indicators

Reusable Components

- Buttons
- Inputs
- Tables
- Loaders

Backend Requirements

APIs

- Attendance APIs
- Task CRUD APIs
- Task assignment APIs
- Comment APIs

Database

- Attendance table
- Tasks table
- TaskAssignments table
- TaskComments table

Deliverables

- Attendance workflow completed
 - Task workflow completed
 - Reusable components created
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Learning Plan (Week 4)

Angular Topics

- Lazy Loading
- Shared Architecture
- Performance Basics
- Search & Filtering
- Pagination

Shared .NET Learning Topics

- Aggregation Queries
- Logging Basics
- API Optimization
- Query Optimization

Assignment 4

Problem Statement

Build Dashboard & Productivity Features.

Dashboard

- Total tasks
- Completed tasks
- Pending tasks
- Attendance percentage
- Employee workload

Charts

- Pie charts
- Bar charts
- Statistics cards

Productivity Features

- Search
- Sorting

- Pagination
- Filters

Notifications

- Notification center
- Read/unread handling
- Badge count

Backend Requirements

APIs

- Dashboard summary APIs
- Notification APIs
- Pagination APIs

Deliverables

- Dashboard completed
 - Charts working
 - Notifications functional
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Learning Plan (Week 5)

Angular Topics

- Environment Configurations
- Optimization Basics
- Better Code Organization
- Error Handling

Shared .NET Learning Topics

- File Handling
- Query Optimization
- Clean Architecture Basics
- Better Validation Handling

Assignment 5

Problem Statement

Build Admin & Production Features.

Admin Module

- User management
- Department management
- Activate/deactivate employees

File Upload

- Attach files to tasks
- File preview support

Activity Logs

- Login activity
- Task creation activity
- Task update activity

Validation Improvements

- Global error handling
- Retry flows
- Better validations

Backend Requirements

APIs

- File upload APIs
- Activity log APIs
- Admin APIs

Deliverables

- Admin module completed
 - File uploads working
 - Activity logs implemented
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Learning Plan (Week 6)

Angular Topics

- Application Stabilization
- Debugging
- API Testing
- Documentation
- Deployment Basics

Shared .NET Learning Topics

- Testing Basics
- Debugging APIs
- Documentation
- Deployment Basics

Assignment 6

Problem Statement

Stabilize the application for internal rollout.

Stabilization

- Fix UI issues
- Fix validation bugs
- Fix API issues

Testing

- API testing
- End-to-end testing
- Validation testing

Documentation

- Setup guide
- API documentation
- Architecture diagram
- User manual

Final Deliverables

- Fully working Angular application
- .NET backend APIs
- SQL database
- GitHub repository
- Documentation
- Architecture diagrams
- Final presentation